MIL-C-85077(AS)
12 December 1977

MILITARY SPECIFICATION

CAPS, BLASTING, SPECIAL

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope. This specification covers electric, shielded blasting caps for use in detonating explosives (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

W-B-411	Blasting-Apparatus (Machines, Blasting; Galvanometers and Rheostats for Testing Blasting Circuits and Machines)
QQ-A-250/8	Aluminum Alloy 5052; Plate and Sheet
QQ-L-201	Lead, Sheet
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-1055	Barrier Material, Waterproofed, Flexible

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department, Code 93, Naval Air Engineering Center, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1375

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Military

MIL-P-387	Pentaerythrite Tetranitrate (PETN)
MIL-P-398	RDX
MIL-G-2550	General Specification for Ammunition Except Small Arms Ammunition
MIL-L-3055	Lead Azide
MIL-C-27500	Cable, Electrical, Shielded and Unshielded, Aerospace

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-1461	Ammunition Manufacturers and Their Symbols

PUBLICATIONS

Department of Defense

4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Materials

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Hazardous Materials Regulations, Department of Transportation

(Copies are available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Material. The materials used in the manufacture of the blasting caps shall be as specified herein. All materials not definitely specified shall be of the best quality normally used for the purpose in good commercial practice. The materials shall be free of all defects and imperfections that might affect the serviceability of the finished product.

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3.1.1 Lead azide. The lead azide shall comply with the requirements of MIL-L-3055.

3.1.1.1 Moisture content of the lead azide. The moisture content of the lead azide, at the time of loading the blasting caps, shall not exceed 0.50 percent, when tested as specified in 4.5.1.

3.1.2 Pentaerythrite tetranitrate (PETN). Pentaerythrite tetranitrate (PETN) shall comply with the requirements of MIL-P-387, class D, except that it shall not be required to pass a vacuum stability test.

3.1.3 RDX. RDX shall comply with the requirements of MIL-R-398, type A or B, class A or B.

3.2 Construction. Each blasting cap shall consist of a shell and a charge. The shell shall be closed at one end.

3.2.1 Shell.

3.2.1.1 Material. The shell shall be made of aluminum alloy in accordance with QQ-A-250/8.

3.2.1.2 Dimensions. The shell shall be of the dimensions specified in table I, when measured (except for length) at a point one-half inch below the open end disregarding any corrugations near the top of the blasting cap.

Table I

DIMENSIONAL REQUIREMENTS

Overall length (inches)	External diameter (inches)	Internal diameter (inches)	Wall thickness (inches)
3.000 ± 0.030	0.271 to 0.279	0.256 to 0.262	0.007 ± 0.002

*NOTE: Maximum diameter of bead of blasting caps shall be 0.0315 inch.

3.2.2 Charge. The charge shall consist of a base charge of not less than 0.94 grams of PETN (see 3.1.2), or of RDX (see 3.1.3), and a priming charge consisting of sufficient lead azide (see 3.1.1) to insure complete detonation of the base charge as required by the performance tests. When necessary, suitable ignition charge shall be used in addition to the priming charge.

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3.3 Details of assembly. The base charge shall be compressed into the shell. A primer charge, or a primer and ignition charge, shall be placed in the shell. The caps shall be provided with a pair of copper lead wires, approximately 24 feet in length, in accordance with the MIL-C-27500, cable designation 22ML2T08, sealed into the cap to make a waterproof assembly. The shielding shall be electrically connected to the cap and to a female connector with a conductive material (solder, braze, or a conductive epoxy such as Emerson and Cuming, Inc., Type 72-G) throughout their respective entire peripheries to form a continuous shield without electrical discontinuities or gaps. The pins on the inner conductors of the connector shall be female and recessed into the insulator to prevent unintentional contact by fingers or tools. In addition, the tips of the female pins shall be recessed well below the lip of the surrounding shield to provide additional shielding. The shielded lead wires shall be covered with a heat shrinkable, flexible, insulation sleeving of appropriate quality. A metal-cased male plug, designed to completely mate with the outer shell (shield) surface before any of its inner shorted conductors make contact with the recessed pins of the connector, when being unmated, shall completely open-circuit the female conductor pins before the shell (shield) contact surface has broken contact. The female connector and plug shall be approved by the contracting officer. The outside of the blasting cap shell shall be legibly marked with the manufacturer's code number per MIL-STD-1461. One male plug with 36 inch leads compatible to female connector, shall be provided with each package of 10 blasting caps for detonation use.

3.4 Functioning.

3.4.1 Electrical resistance. The resistance of the complete cap assembly, including its leads, shall be 2.2 $\pm$ 0.5 ohms when tested as specified in 4.5.3.1.

3.4.2 Waterproofness. The caps shall withstand submersion in water for a period of 4 hours at a temperature of 70 Fahrenheit (F) +5 and a pressure of 50 pounds per square inch (psi) as specified in 4.5.3.2. Following the test, the caps shall function satisfactorily, as evidenced by the perforation of the lead discs when the caps are tested as specified in 4.5.3.2 and 4.5.4.1, and shall either:

(a) Function upon application of an electric current of 1.5 amperes for a period not exceeding 50 milliseconds, as specified in 4.5.3.2, or

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(b) Function through a total resistance of 32 ohms when fired with a blasting machine, as specified in 4.5.3.2.

3.5 Current requirements.

3.5.1 Functioning. The electric caps shall fire when an electric current of 0.45 ampere is applied.

3.5.2 Nonfunctioning. The electric caps shall not fire when an electric current of 0.2 ampere is applied for 5 seconds, as specified in 4.5.5.1.

3.5.3 Firing time. The caps shall fire in accordance with table II.

3.6 Safety. The supplier shall observe the safe practices and standards set forth in DoD 4145.26M.

3.7 Workmanship. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of blasting caps produced by one manufacturer under one contract in one unchanged process and in accordance with the same specification and same specification revision. Each lot shall contain:

(a) Blasting caps

(b) Aluminum shell of new manufacture made by one manufacturer in accordance with the same specification and same specification revision

(c) Lead azide of one manufacturer from the same interfix lot number

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Table II

FIRING TIME VS CURRENT TIME	
Current (D.C.) Amps	Time (Milliseconds)
0.7	10.2
1.0	5.0
2.0	3.0
5.0	2.0

(d) Pentaerythrite tetranitrate (PETN) (where applicable) of one manufacturer from the same interfix lot number

(e) RDX (where applicable) of one manufacturer from the same interfix lot number

(f) Suitable ignition charge (where applicable) consisting of batches blended by the same process, each ingredient being from one lot only

(g) Parts or inert parts made by one manufacturer in accordance with the same specification and same specification revision

4.2.1 Grouping for shipment. When specified in the contract or order, groups for shipment shall be formed of the blasting caps in each lot. The number of units in a group, and the corresponding marking, shall be as specified in the contract or order (see 6.2).

4.3 Sampling. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1 Component materials. The component materials shall be sampled as required for the applicable tests (see 4.5.1).

4.4 Inspection. Inspection shall be as specified in MIL-G-2550.

4.4.1 Dimension inspection. Inspect for compliance with the dimensional requirements in accordance with MIL-STD-105, AQL 2.5%.

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4.5 Test procedures.

4.5.1 Determination of moisture content of lead azide. A tared, dried aluminum dish, 2 inches in diameter, shall be covered with a sheet of polytetrafluoroethylene (see 6.3), 2-1/2 inches square, and sufficiently flat to form a tight cover. An accurately weighed portion of approximately 2 grams of the lead azide shall be transferred to the moisture dish while both the cover and dish area are on the balance pan. The uncovered dish and its contents shall be placed in an oven and dried at 95° centigrade (C) ±2 for 2 hours. The dish cover shall be placed in a desiccator containing anhydrous calcium sulfate, with indicator, during the drying period. After drying, the dish shall be quickly transferred to the disiccator and allowed to cool to room temperature. The polytetrafluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture, as follows:

$$\text{Percentage of moisture} = \frac{(A - B)}{A} \times 100$$

where:

A = weight of the sample before drying, in grams
B = weight of the sample after drying, in grams.

4.5.3 Functioning

4.5.3.1 Electrical resistance. Test sampling shall be in accordance with MIL-STD-105, AQL 1.0%. The electrical resistance of each sample cap, including its leads, shall be measured by the Wheatstone bridge method and meet the requirements of 3.4.1.

4.5.3.2 Waterproofness. Upon completion of the electrical non-functioning electric current test specified in 4.5.5.1, the sample caps shall be conpletely immersed in water for 4 hours at a temperature of 70F ±5, and a pressure of 50 psi. Immediately upon removal from the water, 50% of the sample caps shall be subjected to the static functioning test specified in 4.5.4.1. The remaining 50% shall be either connected in series and an electric current of 1.5 amperes, maximum, applied to the circuit for a period not exceeding 50 milliseconds; or else the caps shall be connected in series, in groups of 10 each, and fired with a 10-cap capacity blasting machine complying with W-B-411, type I, class A, through a total resistance of 32 ohms. Observations shall be made for evidence of complete detonation of the caps to determine compliance with the functioning requirements.

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4.5.3.3 Acceptance. The lot of caps shall be considered acceptable in this phase of testing (electrical resistance and functioning following waterproofness) provided that all the tested caps comply with the applicable functioning requirements and MIL-STD-105, AQL 1.0%.

4.5.4 Static functioning test.

4.5.4.1 Procedure. The lead discs used in this test shall be made of 8-pound sheet lead complying with QQ-L-201, grade B, and shall be 0.1345 inch in thickness and 1.22 -0.01 inches in diameter. The disc shall be centered over a 15/16-inch-diameter hole in a metal support. The cap shall be suitably fixed in a upright position so that the base of the cap is in contact with the lead disc. The caps shall be fired singly, with an electric current of 0.45 amperes, in an upright position, with bottom resting on the unsupported section of the lead disc. Observation shall be made for complete detonation of the cap as evidenced by the perforation of the lead disc.

4.5.4.1.1 Low temperature phase ($-58^{o}F$). Ten sample caps shall be stored at a temperature of $-58^{o}F$, or colder, for a period of 2 hours, minimum. The caps shall then be immediately subjected to the static functioning test specified in 4.5.4. Observation shall be made for perforation of the lead discs to determine compliance with the requirements.

4.5.4.1.2 High temperature phase ($120^{o}F$). Ten sample caps shall be stored at a temperature of $120^{o}F$ for 5 days. The caps shall then be subjected to the static functioning test specified in 4.5.4. Observation shall be make for perforation of the lead disc to determine compliance with the requirements.

4.5.4.2 Acceptance. The lot shall be considered acceptable provided that, in the phase being tested (that is, low temperature phase ($-58^{o}F$), or high temperature phase $120^{o}F$) the caps fire and comply with the applicable static functioning requirements and MIL-STD-105, AQL 1.0%.

4.5.5 Electric current test.

4.5.5.1 Nonfunctioning. The samples selected for 4.5.3.1 shall be subjected to this test. An electric current of 0.20 ampere shall be applied singly to each cap for a period of 5 seconds. Observation shall be made for nonfunctioning of the cap to determine compliance with the requirements.

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4.5.5.2 Acceptance. The lot of blasting caps shall be considered acceptable provided that in the nonfunctioning test the caps comply with the applicable requirements and MIL-STD-105, AQL 1.0%.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit packages. Caps shall have the lead wires laid in folds in such a manner that the folds will open readily without tangling the wires and shall be placed in individual packages 5 inches or more in length, constructed of chipboard having a thickness of not less than 0.010 inch.

5.1.2 Intermediate packages.

5.1.2.1 Unit packages. Unit packages shall be packaged in fiberboard boxes in quantities of 10 unit packages each.

5.1.2.2 Boxes. The boxes shall be snug fitting, W-board interior fiberboard boxes complying with the requirements of PPP-B-636, type SF, class weather-resistant, grade W6s, style FOL, for type 2 load, and closed in accordance with requirements of the appendix thereto.

5.1.2.3 Liners. The unit packages shall be separated with water-resistant fiberboard complying with the requirements of PPP-B-636, grade W6s. Liners shall be used at the sides, ends, top, and bottom of the box, using the same material as specified above, to make a snug fit, if necessary.

5.1.2.4 Marking for identification. Each intermediate box shall be marked or labeled to indicate the item and quantity contained therein as follows:

(10)

NAVAL AIR SYSTEMS COMMAND
ELECTRIC, SHIELDED BLASTING CAPS

No commercial number designation for the cap shall appear on the box.

5.2 Packing.

5.2.1 Intermediate packages. Intermediate packages shall be packed in lined nailed wood boxes in quantities of 1 package each.

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5.2.2 Boxes. Nailed wood boxes shall comply with the requirements of PPP-B-621, class 2, style 2, for type 3 load.

5.2.2.1 Liner. Each shipping container shall contain a barrier material complying with the applicable requirements of PPP-B-1055, classes C1, C2(a), E1, E2, H1, H2, H3(a), H4, H5, L2(b), and M1. The liner shall completely inclose the contents of the box. The dimensions of the liner shall be one-fourth inch greater than the interior length and width of the container to avoid tensional strain when the box is tightly packed. The liner shall be securely sealed at all joints, seams, and folds with a water-resistant adhesive. All flaps shall be folded over, and a rectangular piece of similar liner material equal in size to the width and length of the box, shall be coated with a similar adhesive and securely bonded to the top of the sealed case liner for additional protection.

5.2.2.2 Cushioning. The intermediate packages shall be cushioned with dry excelsior, sawdust, or equivalent material. The cushioning material shall be spread over the bottom of the lined box to a depth of at least 1 inch, and at least 1 inch of the cushioning material shall be packed tightly between the interior containers and the sides and ends of the lined box. In addition all voids shall be filled tightly and the box filled to the level of the top with at least 1 inch of cushioning material to provide a tight pack when the box is closed.

5.2.2.3 Closure and strapping. The wood boxes shall be closed and strapped in accordance with the requirements of PPP-B-621 and the appendix thereto. The straps shall be tensioned and sealed but not stapled.

5.3 Marking. Each wood box shall be marked or labeled to indicate the item and quantity contained therein. The exterior shipping containers shall be marked as required by the Code of Federal Regulations 49 CFR 171-178 and DoD 4145.26M to insure safe handling. In addition marking shall be as specified in MIL-STD-129, except as may be modified by the contracting officer.

6. NOTES

6.1 Intended use. The blasting caps covered by this specification are intended for detonating compressed TNT, compressed nitrostarch, Composition C-3, and Composition C-4 explosives to create earth cavities. These cavities are to be filled with concrete to anchor aircraft and other ground support equipment at expeditionary air fields.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Connector and shorting plug (see 3.3)

(c) Standard resistance of the cap assembly (see 3.4.1)

(d) Grouping for shipment, group size, and corresponding marking, when required (see 4.2.1)

6.3 "Teflon", a product of E. L. du Pont de Nemours and Company, Wilmington, Delaware, has been found satisfactory for making the dish covers used in the moisture content determination of the lead azide (see 4.5.1).

Preparing activity
Navy - AS
(Project No. 1375-N127)

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-85077(AS) Caps, Blasting, Special

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO (If "Yes", in what way?)

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